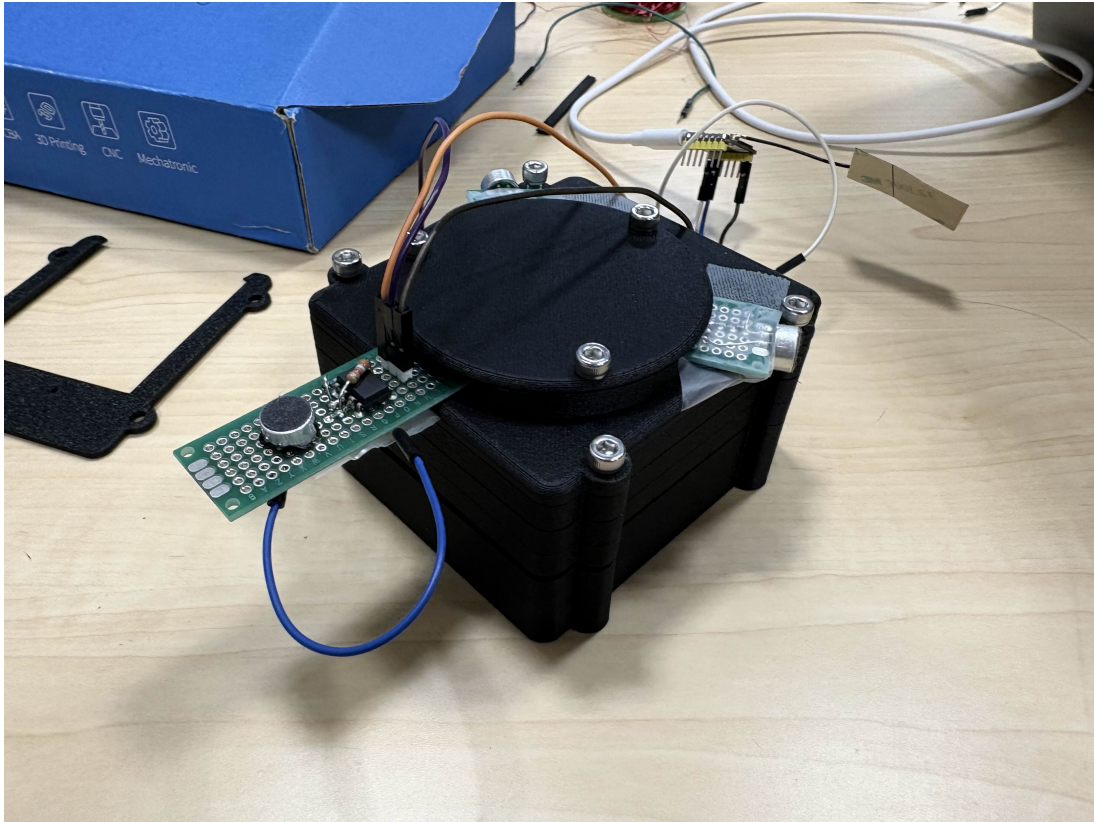


Patrick Liu - WindBorne Firmware Challenge

Firmware Intern application challenge responses | liujpatrick@gmail.com | github.com/machmoon | machmoon.github.io

Cool Thing I Built



One project I liked building was an acoustic sensing grid using an STM32G4 and ESP32 array to localize sound from timing differences. It was fun because it was not just a software problem: the ADC timing, sensor placement, wiring, and signal processing all had to line up. I am proud of it because it made me debug the full system instead of just staring at code.

Setup

I mostly use VS Code/Cursor, a terminal, GitHub, and whatever serial monitor or logging tool the project needs. I like being able to see the actual build output and keep commands repeatable. Over time I have gotten more comfortable working from the terminal because it makes it easier to rerun the same build, flash, and debug steps.

Debugging a PCB That Randomly Resets

First I would try to figure out what kind of reset it is: watchdog, brownout, external reset, hard fault, or something else. My first tests would be checking power with a scope during current spikes, logging the reset reason, watching the reset pin, and trying a known-good supply or cable. If that does not explain it, I would make the failure happen more often with load, temperature, or a firmware feature bisect instead of waiting a few hours each time.

What Firmware/Embedded Engineers Do Wrong

I think embedded people sometimes accept too little visibility into what the system is doing. A board that only fails in the field should still leave clues: reset reasons, counters, logs, and simple test hooks matter a lot. I also do not think abstraction is automatically bad; it is only bad when it hides timing, ownership, or hardware side effects.

Sketchy Firmware Fix

I have definitely added small guard windows or filtering around noisy sensor readings in robotics projects when the demo had to work and the mechanical side was already fixed. It was not the final answer, but it made the system stable enough to test the rest of the project. I am fine with fixes like that when they are clearly marked and do not pretend to be the root-cause fix.

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